

Subject: 23CSE301

Lab Session: 04

Notes & Coding Instructions:

1. Follow the instructions carefully given in this assignment.
2. You are encouraged to use AI tools for code generation and comparison.
3. The prompt log of your interaction should be generated from the tool and submitted.
4. While using GenAI tools, make sure you are doing it from your own login. We shall perform an audit of your interactions with the GenAI tools used.

Main Section (Mandatory):

A1. Please repeat the experiments conducted in Lab 03 (previous lab session) using AI tools. You may use GenAI tools such as Claude, ChatGPT, Co-Pilot. The function definitions and code modularization should be your own work and the AI tools should be used for code generation to meet your needs. Mention the GenAI tool used for each portion of code as comments in the code.

A2. Generate unit test cases using AI tools for testing your modular functions. The unit test cases should be used for both today's and last week's lab exercises.

A3. Conduct a performance comparison of the 2 versions of k-means algorithm, one written by you and the other developed using AI tools.

Report Assignment:

1. Write a report to evaluating the 2 versions of the algorithm developed, one by your own coding and other using AI tools.
2. Report should include observations of functional testing, unit tests conducted and performance tests conducted.
3. Provide the prompt log / conversation history of your interactions with the AI tools. This report should be generated from the AI tool directly.